Notes: Rubinton (RES, 2026)
The Geography of Business Dynamism and Skill-Biased Technical Change

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1 Big picture

- **Question.** Why did large U.S. cities pull away from small cities after 1980 in wages, skill abundance, technology use, and business dynamism?
- **Central mechanism.** A new skill-biased technology is available everywhere, but firms adopt it more in cities where adoption is privately more profitable: large markets, high-skilled amenities, and initial comparative advantage in skilled labour.
- **Technology structure.** The new technology has a high fixed cost and lower marginal cost. It also puts more weight on high-skilled labour. This makes adoption attractive to productive firms and especially attractive in big cities.
- **Selection channel.** More adoption raises wages and rents. Small, low-productivity, non-adopting firms become less profitable and exit. This raises business turnover and amplifies skill-biased technical change.
- **Empirical facts.** In 1980, large and small cities had similar business dynamism. By 2018, large cities had higher start-up and exit rates, while small cities experienced a larger dynamism decline.
- **ICT validation.** Firms in large cities spend more on ICT per employee and devote a larger share of investment to ICT, consistent with higher adoption of new technologies.
- **Quantitative result.** The calibrated model explains essentially all of the increasing city-size relationship in the skill premium and skill intensity, and a little more than half of the increasing relationship in dynamism.
- **Policy lesson.** Policies that raise aggregate technology adoption can amplify geographic inequality; policies that equalize regional outcomes may have weak effects on aggregate adoption.
- **Takeaway.** The Great Divergence is not only about workers sorting into cities. It is also about firms adopting technology and the resulting selection of firms within cities.

2 Data and measurement

2.1 Cities and city size

- The unit of geography is the CBSA/city.
- City size is measured using working-age population, ages 20–64.
- The paper compares cross-sectional relationships with city size between 1980 and 2018.
- The key independent variable in descriptive regressions is $\log(pop_{jt})$, the log working-age population of city j in year t.

Interpretation. City size is the organizing variable because the Great Divergence is fundamentally about the widening gap between large and small urban areas.

2.2 Business dynamism

- Dynamism is measured using establishment start-up rates and establishment exit rates from Business Dynamics Statistics.
- The paper also examines young-firm exit rates because young firms are smaller and closer to the exit threshold.
- The selection hypothesis predicts that, when selection tightens in big cities, low-productivity young establishments should be more affected than old firms.

Interpretation. Dynamism is not treated as merely entrepreneurial energy. In the model, higher turnover can reflect tougher selection: less productive firms exit faster.

2.3 ICT investment

- The paper uses ACES ICT survey data matched to Census firm-level microdata.
- ICT spending includes software, computer equipment, peripherals, maintenance, leasing, and rental payments.
- The main outcomes are log ICT investment per employee and ICT investment as a share of total investment.
- Firm location is assigned using firm establishments; robustness checks use single-establishment firms and manufacturing establishment data.

Interpretation. ICT spending is not a direct binary adoption measure, but it is an observable proxy for firms using the kind of modern skill-biased technology emphasized by the model.

2.4 Wages, skill intensity, and rents

- Wages and skill supply come from the 1980 Decennial Census and 2018 ACS.
- The paper distinguishes high-skilled and low-skilled workers.
- The key regional outcomes are high-skilled wages, low-skilled wages, the skill premium, relative supply of skilled workers, and rents.

Interpretation. The model is disciplined by the full geographic pattern of wages, skill supply, rents, firm dynamics, and ICT usage rather than by one isolated moment.

3 Model and empirical/quantitative specifications

3.1 Descriptive dynamism regression

$$D_{ijt} = \alpha_{it} + \beta_t \log(pop_{jt}) + \varepsilon_{ijt}. \tag{1}$$

- D_{ijt} is a dynamism measure for industry i in city j and year t.
- α_{it} are industry-by-year fixed effects.
- β_t measures the cross-sectional slope of dynamism with city size in each year.

Interpretation. The regression asks whether the relationship between dynamism and city size changes even after controlling for industry composition.

3.2 Final-good demand

$$Y_j = \left(\int_{\Omega_j} q_j(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \qquad q_j(\omega) = Y_j P_j^{\sigma} p_j(\omega)^{-\sigma}. \tag{2}$$

Interpretation. A larger local market raises the scale of demand facing firms. This makes high-fixed-cost, low-marginal-cost technology more attractive in big cities.

3.3 Technology bundles

$$\psi_j^a = \Gamma^\psi \psi_j^n, \qquad \gamma_j^a = \Gamma^\gamma \gamma_j^n, \qquad f_j^a = \Gamma^f f_j^n. \tag{3}$$

- a denotes the adopter technology.
- n denotes the non-adopter technology.
- The new technology is more productive, more skill-biased, and more fixed-cost intensive.

Interpretation. The same new technology is available everywhere. Geographic differences arise because the return to adopting it differs across cities.

3.4 Intermediate production and unit costs

$$q_{jt}^k(\omega) = \psi_j^k z_t(\omega) \left[(1-\gamma_j^k) l_{jt}^{\frac{\epsilon-1}{\epsilon}} + \gamma_j^k h_{jt}^{\frac{\epsilon-1}{\epsilon}}\right]^{\frac{\epsilon}{\epsilon-1}\beta_L} b_{jt}^{1-\beta_L}. \tag{4}$$

$$c_j^k(z) = \frac{C_j^k}{\psi_j^k z}. \tag{5}$$

Interpretation. The key cost object is the city-technology cost index C_j^k. It depends on high- and low-skilled wages and rents, so local labour supply and housing markets shape adoption incentives.

3.5 Adoption threshold

Firms choose the technology that maximizes profits. The marginal adopter is indifferent between old and new technologies. The threshold z_j^a decreases with local output/market size and with the productivity advantage of the new technology, and increases with the fixed-cost gap.

Interpretation. Three forces drive adoption: market size, skill-biased productivity advantage, and fixed-cost burden. Big cities benefit from scale, but only if they can also attract and productively use high-skilled workers.

3.6 Firm dynamics, exit, and selection

$$d\log(z_t) = \mu dt + \Psi dW(t). \tag{6}$$

$$\frac{\partial g_j(s)}{\partial t} = -\widetilde{\mu}\frac{dg_j(s)}{ds} + \frac{1}{2}\widetilde{\psi}^2 \frac{d^2 g_j(s)}{ds^2}. \tag{7}$$

Interpretation. Selection is modeled as an endogenous exit threshold. When adoption raises wages and rents, low-productivity non-adopters become less viable, raising exit and turnover.

3.7 Quantitative experiment

- Calibrate the model to the 1980 cross-section of city fundamentals.
- Introduce a new skill-biased technology plus aggregate changes in the supply of high- and low-skilled labour.
- Compare the new steady state to the 2018 data.
- Evaluate whether the model reproduces changes in wages, skill premium, skill intensity, and dynamism by city size.

Interpretation. The model is not fitted to the cross-sectional changes. Those changes are used as the main quantitative test.

4 Identification and quantitative design

4.1 What identifies the empirical facts

- For dynamism, the key object is the change in the slope of start-up and exit rates with city size from 1980 to 2018.
- For ICT, the key object is the relationship between firm ICT spending and city size, controlling for firm size, age, industry, and year.
- For wages and skill supply, the key object is the changing relationship between city size and skill premium / skill intensity.

Interpretation. The empirical facts are descriptive but carefully chosen to line up with the model's mechanisms: adoption, selection, and skill-biased local demand.

4.2 What identifies the model mechanism

- The model calibrates city fundamentals from 1980, before the main period of divergence.
- The new technology parameters target aggregate wage growth and establishment size, not cross-sectional city slopes.
- The model is then tested by its ability to reproduce cross-sectional changes by city size.

Interpretation. This gives the quantitative exercise a strong discipline: if the model only matched aggregate growth but failed across cities, the adoption mechanism would be weak.

4.3 Policy counterfactuals

- Fixed-cost adoption subsidy: lowers the cost of using the new technology.
- Housing/construction subsidy: expands buildings in above-median population cities, easing rent pressure and shifting population.

Interpretation. These counterfactuals show that aggregate technology policy and place-based equality policy are not the same objective. The model forces policymakers to confront the trade-off.

5 Tables and figures

5.1 Figure 1 + Tables 1–2: Business dynamism and city size

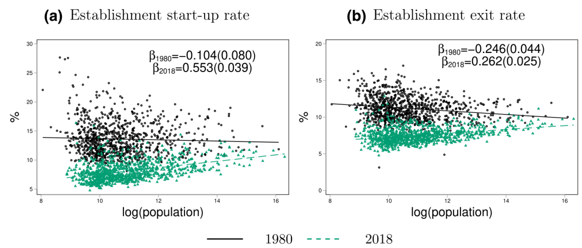

FIGURE 1

Dynamism and city size: (a) establishment start-up rate and (b) establishment exit rate

Notes: Figure shows city-level dynamism, measured by establishment start-up and exit rates, plotted against working-age population (ages 20–64 years). Source: Business Dynamics Statistics and author calculations. Population data come from the Intercensal Population Estimates.

(a) Start-up and exit rates by city size.

TABLE 1

Dynamism and city size, sector controls

	Establishment Start-up Rate		Establishment Exit Rate	
$log(pop_{1980})$	-0.305^{***}		-0.234^{***}	
	(0.0366)		(0.0229)	
$log(pop_{2018})$		0.423^{***}		0.169^{***}
		(0.0208)		(0.0151)
Observations	4534	4534	4534	4534
R^2	0.437	0.607	0.718	0.663

Notes: Results are from a regression of dynamism—measured by establishment start-up and exit rates—on working-age population, controlling for sector fixed effects. Source: Business Dynamics Statistics and author calculations. Working-age population is from the Intercensal Population Estimates. The unit of observation is the CBSA-sector. CBSA-sector pairs with missing dynamism values in either year are excluded.

TABLE 2

Exit rates for young firms and city size

	Establishment Exit Rate: Ages 1–5	
$log(pop_{1980})$	-0.448^{***}	
	(0.0497)	
$log(pop_{2018})$		0.00605
		(0.0351)
Observations	3106	3106
R^2	0.599	0.416

Notes: Results are from a regression of the exit rate for establishments belonging to firms

(b) Sector controls and young-firm exit.

Read: Figure 1 shows a sharp change in the slope of dynamism with city size. In 1980, start-up and exit rates were flat or negatively related to city size; by 2018, large cities have higher start-up and exit rates. Table 1 confirms this pattern after sector controls: start-up slope changes from -0.305 to $+0.423$, and exit slope changes from -0.234 to $+0.169$. Table 2 focuses on young firms: the exit-rate slope for young establishments moves from strongly negative in 1980 to essentially flat in 2018.

Economic message: The aggregate decline in business dynamism hides a regional reallocation. Small cities experience the larger fall in dynamism, while large cities retain or gain relative dynamism.

Additional interpretation: The young-firm evidence is especially important because young establishments are closer to the exit margin. If selection tightens in large cities, young and less productive firms should be most vulnerable. The tables therefore support interpreting dynamism as selection, not just as entrepreneurial entry.

5.2 Table 3: ICT spending and city size

TABLE 3

Technology adoption and city size

	$log\left(\frac{\text{ICT investment}}{\text{employment}}\right)$		$\frac{\text{ICT investment}}{\text{total investment}}$	
log(pop)	0.0519^{***}	0.0554^{***}	0.0116^{***}	0.0119^{***}
	(0.00946)	(0.00737)	(0.00126)	(0.00126)
log(emp)		-0.429^{***}		-0.0120^{***}
		(0.00916)		(0.00167)
log(estabs)		0.479^{***}		-0.00676^{***}
		(0.0115)		(0.00220)
NAICS 4 FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Age FE	N	Y	N	Y
Observations	224,000	224,000	256,000	256,000
R-squared	0.234	0.351	0.172	0.175

SEs clustered at the CBSA level

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: The table reports estimates from regressions of ICT spending on city size. ICT data come from ACES and are merged with the LBD for firm age, size, and location. Working-age population (ages 20–64 years) is from the Intercensal Population Estimates. Note that the difference in sample size between columns 1–2 and 3–4 is due to firms with zero ICT spending.

Table 3: Technology adoption and city size.

Read: City size is positively related to both ICT investment per employee and the ICT share of total investment. The coefficient on log population is about 0.052–0.055 for ICT per employee and about 0.0116–0.0119 for ICT share. The relationship remains after controlling for firm size, age, industry, and year.

Economic message: Larger cities are not simply investing more overall. They appear to allocate more intensively toward ICT, consistent with greater adoption of modern skill-biased technologies.

Additional interpretation: This table is the key bridge between the model's unobserved adoption variable and observable data. ICT spending is not a perfect adoption indicator, but the robust positive relationship with city size validates the model's premise that big cities have more technology adoption.

5.3 Tables 4–5: Calibration of the initial steady state and new technology

TABLE 4

Calibration of initial steady state

Panel A: Aggregate Parameters			
A. Aggregates			
H	Mass of high skill	0.28	data
L	Mass of low skill	1.0	Normalization
B. Elasticities			
$1-\beta$	Housing share of income	0.4	Monte *et al.* (2018)
σ	Elasticity of substitution σ	6.5	Broda and Weinstein (2006)
$1-\beta_L$	Land cost share of production	0.26	Albouy (2016)
ϵ	Elasticity of sub. L & H	1.62	Autor *et al.* (2008)
ν	Shape parameter of Fréchet distribution	1.5	Fajgelbaum *et al.* (2019)
C. Firm productivity process			
$\tilde{\mu}$	Persistence of Brownian motion for s	−0.12	Luttmer (2007)
$\tilde{\psi}$	SD of Brownian motion for s	0.43	Luttmer (2007)

Panel B: City-Level Fundamentals

Parameter	Description of parameter	Description of target	Target
γ_j^n	Initial skill intensity	Skill premium	$\frac{w_{j,h}}{w_{1,l}}$
ψ_j^n	Hicks-neutral productivity	City-size premium	$\frac{w_{j,l}}{w_{j,l}}$
A_{hj}	Amenities, high-skilled workers	Skill intensity	$\frac{H_j}{L_j}$
A_{lj}	Amenities, low-skilled workers	Relative city size	$\frac{H_j+L_j}{H_1+L_1}$
f_j^e	Entry cost	Establishment start-up rate	$\frac{E_j}{M_j}$
f_j^n	Fixed cost	Establishment size	$\frac{H_j+L_j}{M_j}$
ϕ_j	Productivity of building sector	Rent of city j	$\frac{r_j}{w_{1,l}}$
η_j	Elasticity of building sector	Elasticity of building supply	Saiz (2010)
Normalizations			
$\gamma_{1,l}, A_{h1}, A_{l1}$			1

(a) Initial steady-state calibration.

TABLE 5

Calibration of the new technology

Parameter	Description	Value	Target	Data	Model
Γ^{γ}	Skill-bias of new technology	1.90	Aggregate growth in high-skilled wages	28.70	28.71
Γ^{ψ}	Productivity advantage of new technology	1.24	Aggregate growth in low-skilled wages	−3.03	−3.01
Γ^{f}	Additional fixed cost	5.67	Aggregate growth in average establishment size	10.10	10.09
H	Aggregate quantity of high-skilled labour	0.79	Growth in supply of high-skilled labour	0.79	0.79
L	Aggregate quantity of low-skilled labour	1.10	Growth in supply of low-skilled labour	1.10	1.10

(b) New technology calibration.

Read: Table 4 lists aggregate parameters and city-level fundamentals used to rationalize the 1980 data. Table 5 calibrates the new technology: skill bias is 1.90, productivity advantage is 1.24, fixed-cost increase is 5.67, and the model matches targeted aggregate changes in wages and establishment size.

Economic message: The model is disciplined by the 1980 geography before the main divergence and by aggregate post-1980 wage/establishment growth. The cross-city predictions are therefore not mechanically imposed.

Additional interpretation: Table 5 reveals the economic trade-off behind adoption: the new technology is substantially more productive and skill-biased, but its fixed cost is much higher. This is precisely the environment where scale and local high-skill advantages matter.

Read: Figure 2 compares the 1980 data, 2018 data, and 2018 model steady state. The model matches the increasing skill-premium slope and skill-intensity slope almost exactly. Table 6 shows that the baseline explains 116.8% of the cross-sectional change in the skill premium, 101.4% of the change in skill intensity, and 58.6% of the change in start-up rates. Alternative calibrations preserve the qualitative result.

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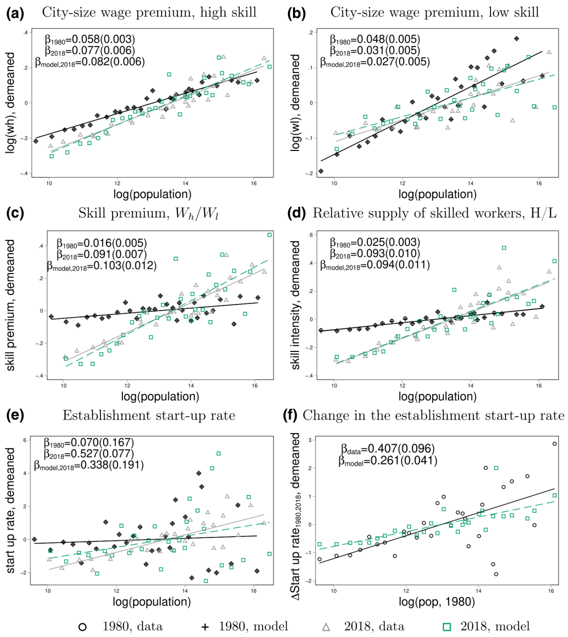

Figure 2

Descriptive facts: data versus model: (a) city-size wage premium, high skill; (b) city-size wage premium, low skill; (c) skill premium, W_h/W_l; (d) relative supply of skilled workers, H/L; (e) establishment start-up rate; and (f) change in the establishment start-up rate

Notes: Source: Wages and skill intensity are from the 1980 Decennial Census and 2018 ACS. Dynamism is from the Business Dynamic

TABLE 6

Robustness to calibration of the new technology

Share of cross-sectional change explained by model in:	Skill-premium	Skill-intensity	Start-up rate
Baseline	116.8	101.4	58.6
Target 50% of wage growth	138.9	116.5	36.4
Allow change in old technology	63.3	81.0	55.1

Notes: Table shows the share of the cross-sectional changes explained by the model. The first row gives the results for the baseline calibration. Rows 2–3 show the robustness of the main results to alternative calibrations.

Economic message: Uneven diffusion of skill-biased technology is powerful enough to explain wage and skill-supply divergence, and selection generated by that diffusion explains a large share of the business-dynamism divergence.

Additional interpretation: The model's weaker fit for start-up dynamism is informative rather than fatal. It suggests that technology-driven selection is a central but incomplete part of the dynamism story; demographic changes, financing conditions, regulation, and local entrepreneurship ecosystems may also matter.

5.4 Figure 3: Drivers of adoption differences

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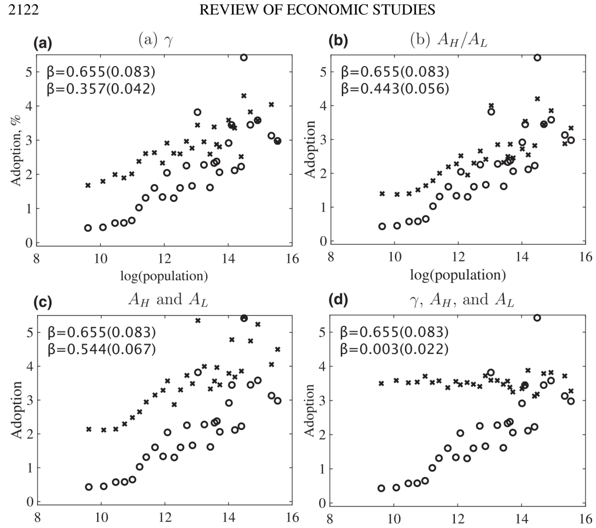

Counterfactual adoption rates.

Read: The baseline adoption slope is 0.655. Equalizing the high-skill weight in initial technology lowers the slope to 0.357. Equalizing relative amenities lowers it to 0.443. Equalizing high- and low-skill amenities lowers it to 0.544. Equalizing all three main sources nearly eliminates the adoption slope, reducing it to 0.003.

Economic message: Adoption is not driven by one factor alone. Market size, initial high-skill productivity, and amenities interact to generate the geography of technology use.

Additional interpretation: Panel (d) is the most important: when multiple city fundamentals are equalized, adoption no longer rises with city size. This suggests that large cities' advantage is endogenous to a bundle of characteristics, not merely population count.

5.5 Figures 4–5 + Table 7: Adoption, selection, and ICT validation

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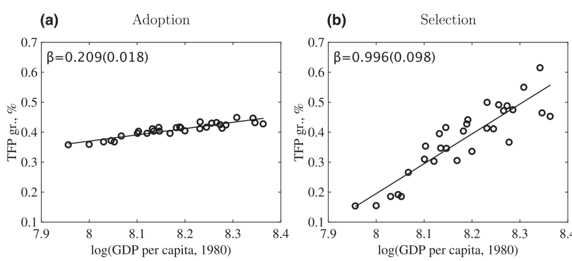

FIGURE 4

TFP growth from adoption versus selection: (a) adoption and (b) selection

Notes: Figure displays the results of a decomposition of annualized TFP growth into growth attributed to adoption and growth attributed

(a) TFP growth from adoption versus selection.

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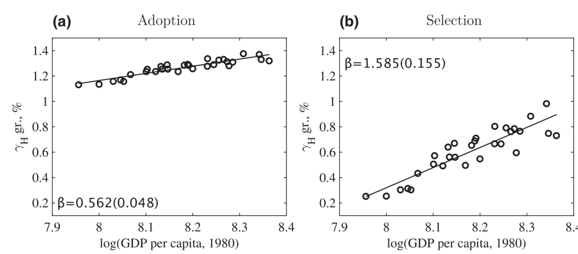


FIGURE 5

Growth in skill-intensity, γ_j, from adoption versus selection: (a) adoption and (b) selection

Notes: Figure displays the results of a decomposition of annualized growth in skill intensity into growth attributed to adoption and growth attributed to selection

(b) Skill-intensity growth from adoption versus selection.

TABLE 7

Model versus Data: ICT spending and adoption versus initial city characteristics

	Data: $log(\frac{ict}{emp})$			Model: share of adopters		
$log(pop_{1980})$	0.0558***			0.655***		
	(0.00708)			(0.0799)		
$\frac{H_{1980}}{L_{1980}}$		2.205***			26.38***	
		(0.427)			(1.085)	
$\frac{w_{h1980}}{w_{l1980}}$			0.313***			11.91***
			(0.105)			(1.178)
Observations	224,000	224,000	224,000	30	30	30
R^2	0.351	0.348	0.347	0.706	0.955	0.236

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Source: The left panel uses ICT data from ACES merged into the LBD for information on the firm's location. Population is working-age population from the Intercensal Population Estimates. The table displays the relationship between the log of ICT spending from 2003 to 2013 and 1980 city characteristics, including city size, skill intensity, and the skill premium. The regressions include a full set of four-digit NAICS, firm age, and year fixed effects. The regressions also control for the log of firm employment and number of establishments. Standard errors are clustered at the CBSA level. The right panel uses model-generated output and shows the relationship between the share of firms that adopt in a city and 1980 city characteristics.

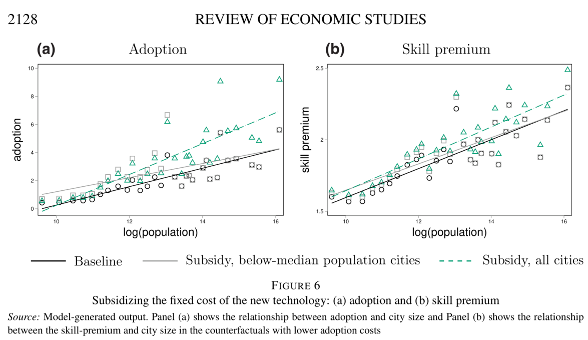
ICT spending and model adoption versus 1980 city characteristics.

Read: Figure 4 decomposes TFP growth: both adoption and selection contribute, but selection is much more strongly related to initial GDP per capita. Figure 5 shows an analogous decomposition for skill intensity: selection has a steeper convergence/divergence slope than adoption. Table 7 validates the mechanisms by showing that ICT spending and model adoption are related to initial population, initial skill intensity, and initial skill premium.

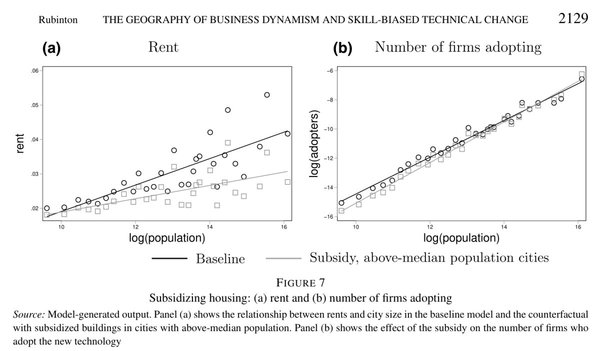
Economic message: Adoption starts the process, but selection amplifies it. Big cities do not only adopt more technology; they also prune low-productivity, less-skill-intensive firms more aggressively.

Additional interpretation: This is the paper's most important mechanism evidence. A pure adoption story would explain why some firms use the new technology, but not why dynamism and productivity distributions change. The selection decomposition shows how firm exit transforms technology adoption into a broader urban productivity and inequality force.

5.6 Figures 6–7: Policy counterfactuals



(a) Subsidizing the fixed cost of the new technology.



(b) Subsidizing housing in large cities.

Read: Figure 6 shows that subsidizing adoption everywhere raises aggregate adoption but increases the skill premium and reinforces geographic differences. Targeting small cities can equalize adoption rates but has little aggregate effect because small cities are small. Figure 7 shows that subsidizing housing in big cities lowers rent dispersion and increases the number of adopting firms, but the share of firms adopting barely rises because lower rents allow marginal non-adopters to survive.

Economic message: Technology policy and place-based equality policy face a hard trade-off. Policies that maximize adoption may deepen geographic divergence, while policies that equalize places may not strongly raise aggregate technology diffusion.

Additional interpretation: The housing counterfactual is subtle: expanding big-city capacity increases the number of adopters through migration and scale, but reduces selection pressure. This means the extensive number of high-tech firms can rise even if the composition of firms does not become much more technology intensive.

6 Short summary

- The paper explains the Great Divergence through uneven adoption of skill-biased technology across cities.
- It documents that large cities are now more dynamic than small cities, unlike in 1980.
- It documents that firms in large cities spend more on ICT per worker and a higher share of investment on ICT.
- The model embeds firm dynamics and technology adoption into a spatial equilibrium with high- and low-skilled workers.
- The new technology is high fixed cost, low marginal cost, and more skill-biased.

- Firms adopt more in large cities because scale, amenities, and initial high-skill productivity increase the return to adoption.
- More adoption raises wages and rents, causing low-productivity non-adopters to exit.
- This selection increases business turnover and amplifies productivity and skill-intensity divergence.
- The calibrated model explains all of the increased skill-premium and skill-intensity slopes and over half of the increased dynamism slope.
- Policy counterfactuals show a trade-off between aggregate technology adoption and regional equalization.

7 Conclusion

7.1 Core conclusion

Rubinton shows that the Great Divergence can be understood as the spatially uneven diffusion of a new skill-biased technology. The technology is available everywhere, but adoption is not uniform because the profitability of paying a high fixed cost differs by city. Large cities, cities that initially use high-skilled labour more productively, and cities with amenities attractive to skilled workers are more likely to adopt.

7.2 Economic intuition

The central intuition is a fixed-cost scale logic embedded in a spatial equilibrium. A firm in a large city can spread the fixed cost of adopting a new technology over a larger market. If the city also attracts skilled workers or is already effective at using skilled labour, the skill-biased technology is even more profitable. Adoption then raises wages and rents, making small non-adopters less viable. The exit of these firms generates selection, raises dynamism, and amplifies the local skill bias.

7.3 What the paper adds to the Great Divergence literature

The paper adds a firm-side mechanism to the worker-sorting and agglomeration explanations of regional inequality. Large cities do not merely attract high-skilled workers; firms in those cities also endogenously choose different technologies. This helps explain why the skill premium, skill supply, ICT intensity, and dynamism all move together with city size.

7.4 Why the selection channel matters

Selection makes the model richer than a simple technology-adoption story. If adoption alone mattered, big cities would have more high-tech firms but not necessarily higher turnover. The paper shows that the adoption shock changes the profitability threshold for survival. Low-productivity firms using the old technology exit, which raises measured dynamism and changes the composition of production. This mechanism helps explain why big cities became more dynamic even as national dynamism declined.

7.5 Policy implications

The policy message is not that technology subsidies are bad. Rather, the model shows that uniform technology subsidies can have non-uniform spatial effects. Because big-city firms have stronger incentives to adopt, a national subsidy may raise adoption most where adoption is already high. Subsidizing small cities can reduce geographic differences, but its aggregate effect is small. Housing subsidies in large cities can increase the number of adopters by permitting more workers and firms to locate there, but they may weaken selection by lowering rent pressure.

7.6 Interpretation for firms and workers

For firms, location changes the return to technology adoption. For workers, technology adoption changes both wages and local firm composition. High-skilled workers gain from cities where adoption is intensive, while low-skilled workers may be hurt both directly by skill bias and indirectly by the exit of firms that are more intensive in low-skilled labour.

7.7 Research agenda

The paper opens several directions. Future work could measure technology adoption more directly, study financing constraints for adoption, separate software from hardware ICT, and explore whether remote work changes the market-size and amenity mechanisms. Another extension would be to examine whether place-based industrial policy can raise adoption without simply shifting population and activity toward already successful cities.

7.8 Takeaway

The paper's main lesson is that new technologies can be nationally available but locally unequal in impact. The interaction of fixed costs, city scale, skilled labour, amenities, rents, and selection can transform a common technology shock into a geographically uneven growth process.